

SOLVED CASE STUDIES

Digital Transformation Tools & Techniques (DTTT)

Final Exam Preparation | 10 Case Studies with Model Answers

Difficulty Breakdown

Level	Count	Case Studies
● Easy	4	CS 1, 2, 3, 4
◆ Medium	4	CS 5, 6, 7, 8
▲ Hard	2	CS 9, 10

Topics Covered

- D365 CRM Overview & Full Customer Journey
 - D365 Sales — Lead Lifecycle, Opportunity, Product Catalog
- Customer Insights – Journeys (Marketing) & Real-Time Marketing
- Customer Insights – Data — Unified Profile, Segments, Predictions
 - D365 Customer Service — Cases, SLAs, Queues, Entitlements
 - D365 Field Service — Work Orders, Scheduling, Mobile App
- Microsoft Power Platform — Canvas Apps, Model-Driven Apps
 - Power Automate, Copilot Studio, Dataverse

Case Study 1 ZestMart Retail — Getting Started with D365 CRM

• EASY

Background

ZestMart is a mid-size Pakistani retail chain with 40 stores. Their sales, marketing, and after-sales teams each use separate spreadsheets and email threads to track customers. The CEO wants to adopt a unified CRM to manage the complete customer journey.

Question 1

Q What is a CRM system, and why does ZestMart need one? Name at least THREE D365 modules that would be relevant for ZestMart.

MODEL ANSWER

A CRM (Customer Relationship Management) system centralises all customer data across an organisation so every department — sales, marketing, and service — works from the same real-time information. ZestMart needs it to eliminate data silos, reduce duplicate records, and give every team a 360-degree view of each customer.

Relevant D365 modules for ZestMart:

- **D365 Sales** — manages leads captured in-store or online, tracks opportunities, generates quotes, processes orders and invoices.
- **D365 Customer Insights – Journeys** — runs automated email campaigns, scores leads by engagement, manages loyalty promotions and event invites.
- **D365 Customer Service** — creates cases for returns, complaints, and warranty claims; enforces SLAs so no customer waits too long.
- **D365 Customer Insights – Data** — consolidates purchase history, website visits, and email engagement into a unified 360-degree customer profile.

Question 2

Q Describe the full D365 customer journey for ZestMart — from first interaction to after-sales support.

MODEL ANSWER

Phase	D365 Tool	What Happens at ZestMart
1. Marketing	Customer Insights – Journeys	Loyalty email campaign targets past buyers. Customer clicks promotional link — interest tracked.
2. Sales	D365 Sales	Interest becomes a Lead. Salesperson qualifies → Opportunity → Quote sent → Order confirmed → Invoice issued.
3. Customer Service	D365 Customer Service	Customer reports a defective product. Case created, routed to relevant queue, SLA timer starts.
4. Field Service	D365 Field Service	Technician dispatched to customer's home for appliance repair. Work Order created and scheduled.
5. Analytics	Customer Insights – Data	All data merged. AI predicts churn risk, recommends retention offers.

Key principle: All D365 apps share the same Dataverse data — no duplicate records, no communication gaps.

Question 3

Q A ZestMart manager says: 'We already use Excel and Gmail — why bother with D365?' Give a precise, terminology-rich counter-argument.

MODEL ANSWER

Excel and Gmail create data silos — each team holds a separate version of customer data with no shared source of truth.

D365 provides: Unified Customer Profile (all data in one place), Timeline (full activity history), Copilot AI (instant summaries and next-action suggestions), Role-Based Access Control (RBAC) via Dataverse, Business Process Flows for consistent cross-team workflows, and real-time SLA tracking for service commitments.

The result: faster decisions, zero duplicate data entry, measurable KPIs, and a scalable platform that grows with ZestMart's operations.

Key Terminology

CRM: System to manage all customer interactions from acquisition to retention

Dataverse: Cloud-based relational database underpinning all D365 apps

360-Degree View: Complete unified customer profile from all data sources

RBAC: Role-Based Access Control — users only see data they are authorised to access

Timeline: History of all activities (emails, calls, meetings, notes) on a record

Case Study 2 TechNova Electronics — D365 Sales Lifecycle

• EASY

Background

TechNova sells enterprise laptops and servers in Pakistan. Their sales team manually tracks deals on paper. They plan to implement D365 Sales and want to understand how the core sales process works.

Question 1

Q Walk through the complete D365 Sales lifecycle for a corporate client who enquired about 50 laptops. Use exact D365 terminology at every stage.

MODEL ANSWER

Stage	D365 Term	What Happens
Enquiry received	Lead	Corporate contact added as Lead. Status: Open. Source: website form.
Research & qualify	Lead Qualification	Salesperson verifies budget, authority, need, timeline (BANT). Lead scored using Predictive Lead Scoring.
Interest confirmed	Opportunity	Lead converted → Opportunity created. Stage: Qualify. BPF (Business Process Flow) activates.
Proposal stage	Opportunity – Propose	Product Catalog used to build Quote. Price List applied. Discount List applied for bulk 50 units.
Client agrees	Order	Customer accepts Quote → Order created in system.
Payment	Invoice	Invoice generated from Order. Final step of sales lifecycle.

Question 2

Q What is the difference between a Qualified Lead and a Disqualified Lead? What happens to each in D365 Sales?

MODEL ANSWER

- **Qualified Lead:** The lead meets all criteria — genuine need, budget available, decision-maker engaged, right timeline. Action: converted to an Opportunity; BPF moves to 'Develop' stage.
- **Disqualified Lead:** No budget, wrong fit, or unresponsive. Action: marked Disqualified in D365 with a reason code. Record retained for future reference but removed from active pipeline.

Question 3

Q TechNova's manager wants to give bulk discounts to government clients but standard retail pricing to others. How does D365 handle this using the Product Catalog?

MODEL ANSWER

D365 Product Catalog supports multiple pricing structures simultaneously:

- Unit Groups — define how a product is sold (1 unit, 1 box of 10, 1-year license).
- Price Lists — Retail Price List (\$1,200/laptop), Government Price List (\$1,050/laptop), Seasonal Sale List (\$1,000/laptop). Each Quote uses the appropriate Price List.
- Discount Lists — additional quantity-based discounts: Buy 1–9 units: 0%, Buy 10–49: 5%, Buy 50+: 10%. Applied automatically when Order quantity qualifies.

When TechNova's salesperson creates a Quote for the government client, they select the Government Price List + Discount List for 50 units — the system calculates the final price automatically.

Key Terminology

BPF: Business Process Flow — Qualify → Develop → Propose → Close guided stages

Lead Scoring: Points assigned to leads based on interactions; high score = sales ready

Product Catalog: System of Unit Groups + Products + Price Lists + Discount Lists

Sales Accelerator: Prioritised work list with sequences — tells salesperson what to do next

Sales Pipeline: All active deals across all BPF stages visible in one view

Case Study 3 FreshBite Food Delivery — Power Platform Adoption

• EASY

Background

FreshBite is a cloud kitchen startup. Their 12 delivery riders currently call the kitchen to update order statuses. The operations manager wants a simple mobile app to let riders update orders and a dashboard for the kitchen manager. No developer is available — only a business analyst.

Question 1

Q Which Power Platform component is most appropriate for FreshBite's mobile rider app, and why? Mention at least TWO other relevant components.

MODEL ANSWER

- **Power Apps (Canvas App)** — most appropriate. Canvas apps provide full UI layout control, work on iOS/Android via Power Apps Mobile, connect to SharePoint or Dataverse, and require no coding — only Excel-like formulas. The analyst can design a rider-facing screen showing assigned orders with an update button.
- **Power Automate** — when a rider marks an order 'Delivered', a flow triggers: updates the data source, sends a confirmation notification to the customer, and logs completion time.
- **Power BI** — kitchen manager's dashboard showing real-time delivery counts, average delivery times, and rider KPIs. Requires Power BI Pro license to share.

Question 2

Q FreshBite stores rider data in a SharePoint list. What are THREE key rules they must follow to make SharePoint work properly with Power Apps?

MODEL ANSWER

1. **Use simple column types only** — Text, Number, Yes/No, Date. Avoid complex column types (Person, Managed Metadata) as they cause errors in Power Apps.
2. **Do not set columns as mandatory in SharePoint** — enforce required fields inside the Power App using form validation, not at the SharePoint column level.
3. **Respect the Delegation Limit** — SharePoint has a threshold for queries returning large datasets. If the order list exceeds this limit, Power Apps shows a delegation warning and may return incomplete results. Solution: filter data on server side using delegable functions.

Bonus: If FreshBite scales to 10,000+ daily orders, they should migrate from SharePoint to Dataverse — which has no delegation issues, supports table relationships, and enables Copilot features.

Question 3

Q What is the difference between a Canvas App and a Model-Driven App? Which is better for FreshBite's rider app?

MODEL ANSWER

Feature	Canvas App	Model-Driven App
Layout control	Full designer control	Determined by data components
Data source	Multiple (SharePoint, Excel, SQL, Dataverse)	Dataverse only
Best for	Custom UI, mobile task apps	Data-heavy back-office apps

Feature	Canvas App	Model-Driven App
Code required	Low-code formulas	Little to no code
FreshBite fit	✓CORRECT CHOICE	✗Not ideal here

Canvas App is correct — FreshBite needs a simple, custom-designed mobile interface for riders, not a data-model-driven back-office app.

Key Terminology

Canvas App: Custom-UI Power App built using drag-and-drop and Excel-like formulas

Gallery Control: Displays a scrollable list of records from a data source

Delegation Limit: SharePoint query threshold — exceeding it returns incomplete data

Power Automate: Workflow automation tool — triggers + actions across apps

Case Study 4 NovaBanking — Internal HR Chatbot

• EASY

Background

NovaBanking's HR team receives 200+ repeated queries per week from employees (leave policy, salary slips, appraisal dates). They want to deploy a chatbot in Microsoft Teams without hiring a developer.

Question 1

Q What tool should NovaBanking use, and what are the exact steps to create and deploy the chatbot to their Teams workspace?

MODEL ANSWER

Tool: Microsoft Copilot Studio (formerly Power Virtual Agents) — creates intelligent conversational agents without coding, deployable directly in Microsoft Teams.

Steps to create and deploy:

4. Open Microsoft Teams → Apps → Search 'Copilot Studio' → Install Microsoft Copilot Studio.
5. Open Copilot Studio → Start Now under 'Create an agent' → Select owning team (HR Team).
6. Provide Agent Name (e.g., 'HR Assistant') and Language. IMPORTANT: Language cannot be changed after creation.
7. Create Topics: Topics page → + New topic → From blank → Add trigger phrases (5–10 per topic, e.g., 'What is the leave policy?', 'How do I apply for leave?').
8. Add message nodes with the HR policy response text. Add Redirect to 'End of Conversation' at the end.
9. Test using the Test Panel on the left side of the authoring canvas.
10. Publish: Copilot Studio → Publish (left menu) → Publish button → confirm.
11. Share: 'Add to a team' — team members can chat privately or @mention in channels.

Question 2

Q Explain Topics, Trigger Nodes, and Conversation Nodes in Copilot Studio with an example from NovaBanking's HR bot.

MODEL ANSWER

Topic: A self-contained conversation unit. Example: 'Leave Policy' topic handles all queries about annual leave, sick leave, and unpaid leave.

Trigger Node: The phrases that activate the topic. NovaBanking should add 5–10 trigger phrases such as: 'leave policy', 'how many leaves do I get', 'annual leave entitlement', 'leave days per year'.

Conversation Nodes — the response logic:

- Show a message: Bot replies with leave policy text.
- Ask a question: Bot asks 'Which leave type? Annual / Sick / Unpaid?' (Multiple choice).
- Add a condition: Branch based on answer — if Annual → show 18 days; if Sick → show 10 days.
- Variables: Store the user's answer (e.g., VarLeaveType = 'Annual') to personalise follow-up messages.
- Built-in variable bot.UserName used to greet: 'Hi Ali, your annual leave entitlement is 18 days.'

Question 3

Q After NovaBanking publishes the bot, what are the three visibility/availability options, and which one requires admin approval?

MODEL ANSWER

- **Copy link** — direct URL shared with specific users only. No admin approval needed.
- **Add to a team** — team members can chat privately or @mention in channels. Visible only to that team. No admin approval needed.
- **Show in Teams app store** → '**Added by your org**' — visible to EVERYONE in the organisation. REQUIRES admin approval before appearing.

Critical note: Any edits to the bot after publishing require republishing before users see changes.

Key Terminology

Topic: Self-contained conversation unit in Copilot Studio with trigger + response nodes

Trigger Node: Phrases/keywords that activate a topic — 5–10 recommended

Conversation Node: Logic step: Show message, Ask question, Add condition, Call action

Variable: Stores user input for use in later nodes (e.g., VarLeaveType)

Case Study 5 GlobeTel Telecom — D365 Customer Service Implementation

◆MEDIUM

Background

GlobeTel is a telecom provider with 3 million subscribers. Their support centre handles 5,000 cases/day. Problems: cases are lost between teams, premium customers get the same wait time as basic customers, and there is no standard process for resolving repeated network complaints. They are implementing D365 Customer Service.

Question 1

Q GlobeTel has Gold, Silver, and Basic tier customers. How should they configure Entitlements and SLAs in D365 Customer Service to reflect different service levels?

MODEL ANSWER

Entitlements define HOW MUCH support a customer receives (support contract). **SLAs** define HOW FAST support must be provided (response and resolution time commitments).

Tier	Entitlement	SLA: First Response By	SLA: Resolve By
Gold	Unlimited phone + chat support, dedicated queue	15 minutes	4 hours
Silver	50 hours phone/year, 100 email tickets/year	1 hour	12 hours
Basic	Email support only, 10 tickets/year	4 hours	48 hours

Configuration steps:

12. Create three Entitlement templates (Gold, Silver, Basic) in D365 Customer Service → Entitlements.
13. Link each Entitlement to the corresponding Account/Contact record.
14. Create three SLA records, each with the First Response By and Resolve By KPIs.
15. Apply the correct SLA to each Entitlement so it activates when a case is created.
16. SLA KPI timers start automatically on case creation and track compliance in real time.

Question 2

Q GlobeTel wants network complaint cases routed to network specialists, billing cases to billing agents, and Gold tier cases always handled first. What D365 routing approach should they use?

MODEL ANSWER

GlobeTel should implement Unified Routing (not Basic Routing) for its intelligence and priority capabilities:

Skills-Based Routing: 'Network' complaint cases → routed to agents with 'Networking' skill tag. 'Billing' cases → routed to agents with 'Billing' skill tag.

Intent-Based / AI Routing: AI analyses case description to classify issue type automatically (e.g., 'My internet is down' → detected as Technical/Network → routed without manual triage).

Capacity-Based Routing: Only routes to agents who currently have capacity — prevents overloading any single agent.

Priority Queues: Gold tier cases assigned higher priority weight so they are picked before Silver and Basic queue items even if created later.

Workstream: A workstream acts as the route manager — it applies classification rules, then routing rules, then optionally tries an AI Virtual Agent first before escalating to a human agent.

Question 3

Q A Gold customer raises 4 separate tickets about the same network outage. How should GlobeTel handle this, and what D365 feature prevents duplicate effort?

MODEL ANSWER

- **Case Merging:** The 4 cases from the same customer about the same outage should be merged into ONE Master Case. The merged cases become child cases linked to the master.
- **Case Hierarchy option:** GlobeTel should configure 'Close All Child Cases' — when the network issue is resolved and the master case is closed, all child cases automatically close. This prevents agents from wasting time closing each ticket individually.
- **Knowledge Base:** Agent resolves the master case → system prompts 'Create knowledge article from this case?' → semi-automatic article created covering the network outage steps. Future similar cases resolved faster using the article.

Key Terminology

Case: Fundamental record representing a single customer service incident

Entitlement: Support contract defining how much support a customer is entitled to

SLA KPI: First Response By + Resolve By — tracked timers on every case

Unified Routing: Advanced routing: skills-based, capacity-based, intent/AI-based

Workstream: Route manager — classifies issue type then decides which agent receives it

Case Merging: Combining duplicate cases into one master case to prevent duplicated effort

Case Study 6 CoolBreeze HVAC — D365 Field Service Implementation

◆ MEDIUM

Background

CoolBreeze installs and services HVAC units for commercial buildings across 3 cities. Their 30 technicians receive job assignments by phone call. Problems: technicians are sent to wrong locations, jobs are double-booked, and customers are not notified before arrival. CoolBreeze is implementing D365 Field Service.

Question 1

Q A customer calls reporting their HVAC unit is leaking. Walk through the complete D365 Field Service process from first contact to invoice, using correct terminology for every step.

MODEL ANSWER

Stage	D365 Action	Details
Customer calls	Work Order Created	Case or direct Work Order raised. Incident Type: 'HVAC Leak' (predefined template — includes required tasks, tools, parts).
Unscheduled	Work Order Status: Open (Unscheduled)	No technician assigned yet. Dispatcher views Schedule Board.
Scheduling	Schedule Assistant used	System suggests best technician based on: HVAC skill tag, proximity to job site (Karachi North zone), current availability (not in another job).
Assigned	Work Order Status: Open (Scheduled)	Technician receives notification on Field Service Mobile App.
En route	Technician uses Mobile App	GPS navigation to customer location. Views customer details, previous service history, required tools.
On site	Status: In Progress	Technician performs repair, records parts used, captures completion data — works offline if no internet.
Done	Status: Completed	Work summary submitted via mobile. System triggers review.
Billing	Invoice Generated	D365 creates invoice from work order — labour, parts, travel all itemised.

Question 2

Q CoolBreeze has 30 technicians across 3 cities. What are the THREE scheduling methods in D365 Field Service? Which is most appropriate for CoolBreeze, and why?

MODEL ANSWER

- **Manual (Schedule Board):** Dispatcher manually drags and drops jobs onto technicians on the Schedule Board dashboard. Full dispatcher control. Best when: small team, complex customer preferences, dispatcher knowledge is critical.
- **Schedule Assistant:** Dispatcher clicks a Work Order → 'Find availability' → system suggests best-matched technicians based on skills, location, and availability. Dispatcher makes final selection.
- **Automated Scheduling (Scheduling Operations Agent):** AI builds an optimised full-day schedule for ALL technicians simultaneously. Considers skills, location, job duration, priority, and travel time. Minimises mileage and maximises jobs per technician per day.

✓Best for CoolBreeze: Schedule Assistant — they have 30 technicians across 3 cities and need skills-based matching (HVAC installation vs repair) but still want dispatcher oversight. Full automation (Scheduling Operations Agent) is ideal if they scale to 100+ technicians.

Question 3

Q What is an Incident Type in Field Service, and how does it help CoolBreeze standardise their most common job: 'Annual HVAC Maintenance'?

MODEL ANSWER

An Incident Type is a predefined job template for a recurring type of work. Instead of manually adding tasks and tools every time, the dispatcher selects the Incident Type and the system populates everything automatically.

CoolBreeze's 'Annual HVAC Maintenance' Incident Type would include:

Tasks: (1) Inspect filters, (2) Clean condenser coils, (3) Check refrigerant levels, (4) Test thermostat calibration.

Required Tools: Filter cleaning kit, refrigerant gauge, vacuum pump.

Estimated Duration: 2.5 hours.

Required Skill: HVAC Certified Technician.

Benefit: Every technician follows the same standardised process — consistent service quality, no missed steps, faster job creation, and accurate scheduling (2.5-hour slot booked automatically).

Key Terminology

Work Order: The core job record in Field Service — lifecycle:

Create→Schedule→Dispatch→Service→Review→Invoice

Incident Type: Predefined template for common jobs — includes tasks, tools, skills, duration

Schedule Board: Dispatcher's main dashboard — visualises technicians, jobs, time slots, and map

Schedule Assistant: Automated technician suggestion tool based on skills + location + availability

First-Time Fix Rate: KPI measuring % of jobs resolved on the first technician visit

Case Study 7 StyleHub Fashion — Marketing & Customer Insights

◆MEDIUM

Background

StyleHub is an online fashion retailer. Their marketing team sends the same promotional email to all 500,000 subscribers. Result: 3% open rate, high unsubscribe rate. The CMO wants to implement D365 Customer Insights (Journeys + Data) to deliver personalised, trigger-based marketing campaigns.

Question 1

Q StyleHub wants to automatically send a personalised discount email when a customer adds items to their cart but does not purchase within 2 hours. Which D365 feature handles this, and how?

MODEL ANSWER

This is Real-Time Marketing using a Customer Journey with a Custom Trigger.

17. **Segment (WHO):** Create a Dynamic Segment — 'Customers who added to cart but did not purchase in last 2 hours.' This segment auto-updates every time the condition is met.
18. **Trigger (WHEN):** Custom Trigger set on cart-abandonment event ('CartAbandoned' event fires when no purchase within 2 hours). Real-Time Marketing responds INSTANTLY when the trigger fires.
19. **Marketing Activity (WHAT):** Personalised Email: 'Hi [bot.UserDisplayName], you left something behind! Here's 10% off to complete your order.' — Dynamic Content shows the specific cart items.
20. **Journey flow:** Trigger fires → Enter Journey → Wait 2 hours → Send personalised email → If customer clicks: route to purchase confirmation → If no action after 24 hours: send SMS reminder.

Exam Tip — Always remember the 3 elements of any Customer Journey: WHO (Segment) + WHAT (Email/Content) + WHEN (Trigger)

Question 2

Q StyleHub wants to create two customer groups: (1) Customers who attended their 2023 year-end sale event, and (2) Customers who spent PKR 50,000+ in the last 3 months. Which segment type applies to each, and why?

MODEL ANSWER

- **Group 1 → Static Segment:** 'Customers who attended the 2023 year-end sale event' is a fixed, past event. The list does not change — new customers cannot retroactively attend. Static segments are created once and do not auto-update.
- **Group 2 → Dynamic Segment:** 'Customers who spent PKR 50,000+ in the last 3 months' changes continuously. New qualifying customers are auto-added; customers who drop below the threshold are auto-removed. Perfect for ongoing high-value targeting.

Question 3

Q How does D365 Customer Insights – Data build a Unified Customer Profile for StyleHub? What are Measures, Segments, and Predictions in this context?

MODEL ANSWER

Unified Customer Profile: All data from different sources is collected (ingestion), matched (same email = same person), and merged into ONE 360-degree customer profile.

StyleHub data sources: website visits (Behavioral Data), purchase history (Transactional Data), demographic information (age, city, gender) (Demographic Data). Combined → one profile per customer.

Measures (KPIs): Quantified business metrics. For StyleHub: Total Lifetime Spend per customer, Average Order Value, Number of Returns. Used to identify high-value vs low-value customers.

Segments: Groups of similar customers. StyleHub examples: Dynamic Segment 'Customers spending 50K+ in 3 months'; Static Segment 'VIP event attendees'. Sent to D365 Journeys for targeted campaigns.

Predictions (AI): Churn Prediction — AI flags customers likely to stop buying (so StyleHub sends a retention offer before they leave). Product Recommendation — AI suggests next purchase. Customer Lifetime Value — AI estimates future revenue from each customer.

Key Terminology

Real-Time Marketing: Instant response to customer actions — trigger fires, journey starts immediately

Dynamic Segment: Auto-updating customer group based on live data conditions

Static Segment: Fixed list that does not change automatically — past event attendees

Customer Journey: Automated sequence: WHO (segment) + WHAT (content) + WHEN (trigger)

Churn Prediction: AI model identifying customers at risk of stopping purchases

Unified Customer Profile: 360-degree view — all data sources merged into one customer record

Case Study 8 HealthFirst Clinic — Model-Driven App for Patient Management

◆MEDIUM

Background

HealthFirst Clinic manages 10,000 patient records. Doctors need to view and update patient histories; receptionists need to create and schedule appointments; billing staff need to view invoices only. They are building a Model-Driven App in Power Apps backed by Dataverse.

Question 1

Q Why is a Model-Driven App more appropriate than a Canvas App for HealthFirst? Describe the 5 phases of building the Model-Driven App.

MODEL ANSWER

Model-Driven App is correct because: HealthFirst data is complex with multiple related tables (Patients, Appointments, Doctors, Invoices, Medical Records). Model-Driven apps are metadata-driven — the layout is automatically generated from the Dataverse data model, including relationships, views, and forms. No custom UI design needed for data-heavy back-office workflows.

Phase	Action	HealthFirst Example
1. Model Business Data	Build Dataverse tables and relationships first	Tables: Patient, Appointment, Doctor, Invoice. Relationships: 1 Patient → Many Appointments (1:N)
2. Define Business Processes	Create Business Process Flows	Appointment BPF: Schedule → Confirm → Check-in → Consult → Follow-up
3. Compose the App	Use App Designer — select pages	Add Patient form, Appointment view, Doctor lookup. Power Apps auto-creates site map.
4. Configure Security Roles	Assign privilege levels per table per role	Doctor: Read + Write on Patient + Medical Records. Receptionist: Create + Read on Appointments.
5. Share the App	Assign security role THEN share app	Two-step: (1) assign role → (2) share. Sharing without security role = user cannot access data.

Question 2

Q Configure the exact security privileges for each of HealthFirst's three user roles: Doctor, Receptionist, and Billing Staff. Use the correct D365 privilege terms.

MODEL ANSWER

Privilege	Doctor	Receptionist	Billing Staff
Create	✓Medical Records	✓Appointments	✗
Read	✓All patient data	✓Appointments + Patient names	✓Invoices only
Write	✓Medical Records, Patient history	✓Appointments	✗
Delete	✗	✗	✗
Append	✓Attach notes to patient record	✓Attach appointment to patient	✗
Append To	✓Add notes to patient	✗	✗

Privilege	Doctor	Receptionist	Billing Staff
Assign	X	✓Assign appointments to doctors	X
Share	X	X	X

Key exam note: Append = associate current record TO another. Append To = allow another record to be associated WITH the current record. These are frequently confused in exams.

Question 3

Q HealthFirst's admin changes a Patient table column from 'Text' to 'Number' type. What will happen, and what is the correct procedure?

MODEL ANSWER

✓CRITICAL: In Dataverse / Model-Driven Apps, column types CANNOT be changed after creation. Attempting to change the type (e.g., 'Phone Number' column from Text to Number) requires: DELETE the existing column (ALL data in that column is permanently lost) → Recreate the column with the correct data type → Re-enter or re-import the data.

Correct procedure for HealthFirst: (1) Export all existing data from the column to Excel. (2) Delete the column. (3) Create a new column with the correct type. (4) Re-import the data.

Prevention: Always validate column types with stakeholders before going live. Changing a column type is a data-loss risk — flag this in every design review.

Key Terminology

Model-Driven App: Data-first app — layout determined by Dataverse components, not the designer

Metadata-Driven Architecture: App design based on data structure — no code needed when data model changes

Append vs Append To: Append: associate current record to another. Append To: allow others to associate to current.

Business Process Flow: Guided step-by-step process — can span up to 5 tables, up to 10 active per table

Security Role: Set of privileges per table — must be assigned before user can access app data

Case Study 9 MegaBank — Full D365 Digital Transformation

▲HARD

Background

MegaBank Pakistan has 2 million customers and 5,000 employees across 300 branches. Their current IT landscape: a legacy core banking system (IBM), separate Excel-based leads tracking, a third-party call centre tool, manual field visit logs, and disconnected marketing emails. The CTO wants to implement a full Microsoft Dynamics 365 + Power Platform solution to unify operations. The board has approved a 3-year transformation roadmap.

Question 1

Q Map EVERY D365 and Power Platform component MegaBank needs, justify each choice, and explain how they integrate via Dataverse.

MODEL ANSWER

Component	Justification	Integration Point
D365 Sales	Replace Excel leads tracking. Full Lead → Opportunity → Quote → Order → Invoice lifecycle for banking products (loans, credit cards, investment accounts).	All lead/opportunity data stored in Dataverse. Shared with Customer Insights – Data for profile enrichment.
Customer Insights – Journeys	Automated email/SMS campaigns for product promotions. Real-Time Marketing triggers when customers open accounts or qualify for upgrades.	Segments fed from Customer Insights – Data. Behavioral data (email opens, link clicks) written back to Dataverse.
Customer Insights – Data	Unify all customer data: transactions, loan history, branch visits, call centre logs. 360-degree Unified Customer Profile. Churn Prediction for at-risk customers.	Ingests from core banking (IBM via connector), D365 Sales, Customer Service, and Power Pages portal.
D365 Customer Service	Handle customer complaints, fraud reports, loan disputes as structured Cases. SLAs per customer tier (Priority/Standard). Knowledge Base for agents.	Cases linked to customer Accounts in Dataverse. Queues route to correct team (Fraud, Loans, Cards).
D365 Field Service	Schedule relationship manager visits to HNW (High Net Worth) clients. Work Orders for ATM maintenance.	Work Orders in Dataverse. Technician locations synced via Mobile App.
Power Apps (Canvas)	Branch staff mobile app: view daily task list, check customer appointment schedule, update visit outcomes offline.	Reads/writes to Dataverse tables via connectors.
Power Apps (Model-Driven)	Relationship Manager back-office app: full customer 360 view, activity history, linked opportunities and cases.	Metadata-driven from Dataverse Customer/Opportunity/Case tables.
Power Automate	Automate: new account email welcome, fraud alert notifications,	Triggers from Dataverse row changes, D365 events, and email (Office 365 connector).

Component	Justification	Integration Point
	loan approval workflow (multi-step approval chain), SLA breach escalation.	
Copilot Studio	HR chatbot in Teams (leave, payslip queries). Customer-facing chatbot on Power Pages (balance enquiry, statement download).	HR bot reads from Dataverse HR tables. Customer bot integrates with Customer Service via handoff to human agent.
Power BI	Executive dashboards: loan disbursement vs target, customer churn rate, SLA compliance, sales pipeline health. Embedded in Model-Driven App.	Power BI Pro required. Direct query from Dataverse managed data lake.
Power Pages	External customer portal: apply for loan, track application status, download statements, raise service requests.	Authenticates via Azure AD. Reads/writes to Dataverse. Raises Cases in D365 Customer Service automatically.

Dataverse as the integration backbone: Every D365 module and Power Platform component writes to and reads from Dataverse. This eliminates data silos — the same customer record is visible to Sales, Service, Marketing, and Analytics simultaneously. RBAC ensures branch staff see only their customers; relationship managers see their portfolio; executives see aggregated dashboards.

Question 2

Q MegaBank's risk team is concerned about data governance. How does the Power Platform Admin Center address this? Cover environments, data policies, and security roles precisely.

MODEL ANSWER

Power Platform Admin Center (admin.powerplatform.microsoft.com) gives MegaBank's IT team centralised governance:

Environments: MegaBank should create separate environments: Development (build and test new apps), UAT/Sandbox (non-production testing, isolated from production data), Production (live apps and data). This supports ALM (Application Lifecycle Management): Dev → UAT → Production.

Data Loss Prevention (DLP) Policies: Admin creates connector policies that classify connectors as Business (approved: SharePoint, Dataverse, Teams) or Non-Business (blocked: Twitter, personal Gmail). This prevents employees from building flows that send customer data to unapproved external services.

Security Roles in Dataverse: Every table in Dataverse has granular security privileges (Create, Read, Write, Delete, Append, Append To, Assign, Share). MegaBank assigns: Relationship Managers → Read+Write on Customer + Opportunity tables. Branch Tellers → Read only on Customer table. Fraud Team → Read+Write+Delete on Case (Fraud) table. Executive → Read on all tables (reporting only).

Changes in the Admin Center are immediately available to all users — no republishing needed for policy changes.

Question 3 — Tricky

Q MegaBank's Copilot Studio chatbot was built in English for Pakistan operations. Six months later, they expand to Saudi Arabia and need the same bot in Arabic. The developer says 'we can just change the language setting.' Is this correct? What is the right approach?

MODEL ANSWER

✓The developer is **WRONG**. In Copilot Studio, the agent language **CANNOT** be changed after creation. This is a hard platform limitation.

Correct approach: Create a brand-new Copilot Studio agent for the Saudi Arabia deployment, select Arabic during creation (before clicking Create — this setting is permanent).

The English bot and Arabic bot must be maintained as two separate agents. Content (topics, trigger phrases, responses) must be re-created in Arabic in the new agent.

Strategic recommendation: For multi-language deployments, create one agent per language from the start of the project — not as an afterthought. Trigger phrases and conversation nodes in Arabic must reflect natural Arabic banking vocabulary, not translated English phrases.

Key Terminology

ALM: Application Lifecycle Management — Dev → UAT → Production environment pipeline

DLP Policy: Data Loss Prevention — classifies connectors as Business (allowed) or Non-Business (blocked)

Managed Environments: Secure, isolated Power Platform environments with enhanced governance controls

Fusion Development: Citizen developers (power users) + professional developers working together on the same platform

Unified Customer Profile: 360-degree view built by Customer Insights – Data by merging all data sources

Case Study 10 EduReach University — End-to-End CRM Transformation

▲HARD

Background

EduReach University has 25,000 students, 400 faculty, and 3 campuses. They are experiencing: low student retention (students drop out without early warning), poor prospective student follow-up (leads go cold), no automated marketing for course promotions, inefficient exam complaint resolution (30-day average), and no data-driven understanding of student satisfaction. The CTO proposes a full Microsoft Dynamics 365 + Power Platform implementation treating students as 'customers'.

Question 1 — Analyse & Design

Q Map each university problem to the precise D365/Power Platform solution. For each, name the exact feature/module, explain the configuration, and describe the measurable outcome.

MODEL ANSWER

Problem	D365/PP Solution	Configuration	Measurable Outcome
Low student retention — no early warning	Customer Insights — Data: Churn Prediction	Train AI model on historical dropout data. Factors: attendance (Behavioral), grade drops (Transactional), fee delays (Transactional). Students scoring high churn risk → Dynamic Segment auto-created.	Retention team alerted 60 days before risk point. Target: 15% reduction in dropouts.
Leads going cold — prospective students	D365 Sales: Lead → Opportunity pipeline + Sales Accelerator	Inquiry form submission = Lead created. BPF: Qualify (verify interest) → Develop (campus tour scheduled) → Propose (offer letter) → Close (enrollment confirmed). Sales Accelerator work list tells admissions what to do each day.	Lead response time from 5 days to same-day. Conversion rate target: +20%.
No automated marketing	Customer Insights — Journeys: Real-Time Marketing	Trigger: student completes online inquiry form → Journey starts → Welcome email (personalised with program name) → Day 3: SMS reminder → Day 7: webinar invite → Day 14: offer letter email. Segments: by program interest (IT, Business, Medicine).	Open rate target: from 3% to 25%. Unsubscribe rate reduced by dynamic segment personalisation.
30-day average exam complaint resolution	D365 Customer Service: Cases + SLAs + Queues	Case created per complaint. SLA: First Response By = 24 hours, Resolve By = 5 working days. Queue: Exam Complaints queue → routed to Academic Affairs team. Entitlement: all students entitled to 3 formal complaints per semester.	Resolution time: 30 days → 5 working days. SLA KPI tracked in Power BI dashboard.

Problem	D365/PP Solution	Configuration	Measurable Outcome
No student satisfaction data	Customer Insights – Data: Measures + Copilot Studio survey bot	Measures configured: average satisfaction score, NPS per faculty. Copilot Studio bot in Teams sends automated satisfaction surveys post-exam, post-course. Results written to Dataverse. Power BI dashboard shows faculty rankings.	First satisfaction baseline established in semester 1. Low-scoring faculty flagged for review.

Question 2 — Tricky Scenario

Q The Academic Affairs head says: 'We should build a Canvas App for the exam complaint system so we can fully customise the UI.' The IT manager says: 'We should build a Model-Driven App because all complaint data is in Dataverse.' Who is correct, and what is the nuanced answer?

MODEL ANSWER

The IT manager is more correct for the core complaint management system — but the full answer is nuanced.

Why Model-Driven App is correct for complaint management:

- All complaint data (Cases) is already in Dataverse as part of D365 Customer Service.
- A Model-Driven App is metadata-driven — forms, views, and relationships are auto-generated from the Case table structure. No custom UI design needed.
- Business Process Flows (Complaint Received → Under Review → Decision Made → Resolved) are only available in Model-Driven Apps.
- Security roles on the Case table already control who can create/read/resolve complaints.

Why Canvas App could supplement (not replace) it:

- A student-facing Canvas App could serve as the submission portal — simple, branded, mobile-friendly. Student submits complaint → Power Automate creates a Case in D365 Customer Service automatically.
- This is the correct architectural pattern: Canvas App (student-facing front end) → Power Automate (creates record) → D365 Customer Service Model-Driven App (back-office processing).

Exam answer: Both are needed — Canvas App for student submission (custom UI), Model-Driven App for staff processing (data-driven, BPF, security roles). This is Fusion Development in practice.

Question 3 — Most Tricky

Q EduReach's Power Automate admin is told: 'Build a flow that saves all student inquiry emails from Outlook to SharePoint AND posts a summary to the Admissions Teams channel AND creates a Lead in D365 Sales — all automatically.' The admin says it's impossible with one flow. Is this correct?

MODEL ANSWER

✓The admin is WRONG. This is a single Power Automate Cloud Flow with one Trigger and three Actions. Trigger: Office 365 Outlook — 'When a new email arrives' in the Admissions inbox (filter by subject keyword 'course inquiry').
Action 1: SharePoint — 'Create file' — saves the email body as a .txt file to the 'Student Inquiries' document library.

Action 2: Microsoft Teams — 'Post a message in a channel' — posts a formatted summary card to the Admissions channel with sender name, email, and program of interest.

Action 3: D365 Sales — 'Create a new record' — creates a Lead record with: First Name, Last Name, Email, Lead Source = 'Email', Description = email body.

This is a standard multi-action cloud flow. Power Automate supports unlimited actions in a single flow. The 600 flows/account limit refers to the NUMBER of flows, not actions within a flow.

Additional note: This flow can be moved between Development → Production environments using Solutions (the preferred method) — not zip export, which is legacy.

Key Terminology

Churn Prediction: AI model identifying at-risk customers/students before they disengage

BPF (Business Process Flow): Guided multi-stage process — available only in Model-Driven Apps — up to 5 tables, 10 per table

Fusion Development: Canvas App (custom UI) + Model-Driven App (data logic) working together — citizen + pro developer collaboration

Solutions: Preferred method for moving Power Automate flows between environments (not zip export)

Trigger vs Action: Trigger: event that starts the flow. Action: what happens in response. One trigger, unlimited actions.

Real-Time Marketing: D365 Journeys feature — responds instantly to customer actions using trigger-based journeys

End of Solved Case Studies • Best of luck, Shariq!